

# Pricing Arithmetic Asian Options using Monte Carlo Simulation

Prepared by **Gautam Prasad & Monica Bernatowicz**

Live demo: [asianoptionprice.streamlit.app](https://asianoptionprice.streamlit.app)

---

## 1. Introduction

This report documents the implementation and analysis of a pricing engine for the arithmetic Asian call option. An interactive version of the engine is hosted at the link on the cover page and accepts the full set of input parameters (spot, strike, volatility, risk-free rate, maturity, monitoring frequency, sample size, and option type).

The report covers the following:

- Definition of the instrument and the economic motivation for trading it.
- Pricing methodology, including the choice of variance reduction technique.
- Numerical results for a representative example.
- Sensitivity analysis with respect to each input parameter.

A single numerical example is used throughout the report, corresponding to the default configuration of the engine.

### OUR RUNNING EXAMPLE

Inputs:  $S_0 = 100$ ,  $K = 100$ ,  $\sigma = 20\%$ ,  $r = 5\%$ ,  $T = 1$  year,  $m = 252$  (daily),  $N = 50,000$ , Call.

Result: **Option price = 5.7828**    95% confidence interval = [5.7807, 5.7848].

## 2. Definition of the Asian option

The most widely known option contract is the European option. A European call grants the holder the right to buy an underlying asset at a fixed strike price on a single specified future date. The payoff therefore depends exclusively on the terminal price of the underlying.

Asian options modify this construction in a single but consequential way: the payoff depends on the average price of the underlying over a specified monitoring window, rather than on its closing price. To illustrate, consider a hypothetical underlying that begins at 100, rises to 150 mid-year, and finishes the year at 90. A European call on this underlying observes only the terminal value of 90, whereas an Asian call evaluates the average over the entire path.

The payoff of an arithmetic Asian call at maturity is:

$$\text{Payoff} = \max(\bar{A} - K, 0)$$

Where:

- $\bar{A}$  is the average price of the underlying over the monitoring window
- $K$  is the strike price (the price agreed in the contract)
- $\max(x, 0)$  is the larger of  $x$  and zero. If the average ends up below the strike, the payoff is zero, because no one would choose to exercise at a loss.

The average  $\bar{A}$  is built from  $m$  equally spaced observations:

$$\bar{A} = \frac{1}{m} (S(t_1) + S(t_2) + \cdots + S(t_m))$$

Where  $S(t_i)$  is the asset price recorded on observation date  $t_i$ . In our example  $m = 252$ , which means we sample the price once per trading day over the year.

The corresponding Asian put has payoff  $\max(K - \bar{A}, 0)$ . Both variants are supported by the engine.

### 3. Practical use of Asian options

Asian options occupy a central role in the over-the-counter market for commodities and foreign exchange. The reason is structural: for many corporate users, the underlying economic exposure is to an average price rather than to a single closing price.

A representative example is an oil refinery that purchases approximately equal volumes of crude oil on each trading day of the year. The refinery's annual procurement cost therefore depends on the average daily oil price, not on the closing price on the final trading day. To cap this annual cost, the refinery can purchase an Asian call on crude oil with strike equal to its budgeted average price. If the realised average exceeds the strike, the option payoff offsets the additional procurement cost. A European call written on the closing price would constitute a mismatched hedge, since the firm's exposure is distributed over the full year rather than concentrated on a single date.

The same logic applies to a European exporter receiving US dollar revenues over a quarter, to an index fund whose performance is referenced to a monthly average, and to many other corporate hedging programmes. Whenever the underlying exposure is to an averaged price, the Asian option provides the corresponding tailored hedge.

### 4. Limitations of the Black-Scholes framework

The standard Black-Scholes formula applies to European options because the underlying price on a single future date is lognormally distributed under the model. The expected payoff can therefore be written in closed form using  $d_1$ ,  $d_2$  and the normal cumulative distribution function.

For an arithmetic Asian option this approach breaks down. The relevant quantity is the expected value of the maximum of an arithmetic *average* of correlated lognormals. While each individual price  $S(t_i)$  is lognormal, the sum of correlated lognormals is not. Since the Black-Scholes derivation relies critically on lognormality, no closed-form analogue exists for the arithmetic Asian. The standard solution adopted in both academic and practitioner work, and the one implemented in the engine, is Monte Carlo simulation.

## 5. Pricing methodology

### 5.1 Model of the underlying

The engine adopts the same risk-neutral Geometric Brownian Motion that underlies the Black-Scholes model:

$$dS(t) = r S(t) dt + \sigma S(t) dW(t)$$

Where:

- $S(t)$  is the asset price at time  $t$
- $r$  is the risk-free rate (5% per year in our example)
- $\sigma$  is the volatility (20% per year in our example)
- $W(t)$  is a Brownian motion, which can be thought of as the cumulative random noise driving the price.

This stochastic differential equation has a clean closed-form solution:

$$S(t) = S_0 \exp\left((r - \frac{1}{2}\sigma^2)t + \sigma W(t)\right)$$

In simulation, we sample one Brownian increment for each small time step  $\Delta t = T/m$ . With  $T = 1$  year and  $m = 252$ , the step size is one trading day.

## 5.2 Monte Carlo estimator

Under the risk-neutral framework, the fair price of an option today is the expected payoff discounted back to today at the risk-free rate:

$$V_0 = e^{-rT} \mathbb{E}[\max(\bar{A} - K, 0)]$$

Where:

- $V_0$  is the option price today
- $e^{-rT}$  is the discount factor that brings money received at maturity back to today
- $\mathbb{E}[\cdot]$  is the expected value under the risk-neutral measure

The Monte Carlo estimator approximates this expectation in three steps:

1. Simulate  $N$  independent price paths under the model above.
2. On each path, compute the average  $\bar{A}$  and the payoff  $\max(\bar{A} - K, 0)$ .
3. Average the discounted payoffs across all  $N$  paths to obtain the estimate of  $V_0$ .

The estimate becomes more accurate as  $N$  grows: its standard error shrinks at the rate  $1/\sqrt{N}$ . Halving the error therefore requires quadrupling the path count, and reducing it by an order of magnitude requires a hundredfold increase. This slow convergence is the main weakness of plain Monte Carlo and motivates the variance reduction technique introduced next.

## 5.3 Limitations of plain Monte Carlo

Applying the plain Monte Carlo estimator to the running example with  $N = 50,000$  paths produces a price estimate of approximately 5.77 with a standard error of 0.036. The associated 95% confidence band spans roughly  $[5.70, 5.84]$ , which for a contract worth less than six units is too wide to support a confident quote. Tightening the band by an order of magnitude through brute force alone would require a hundredfold increase in the number of paths, which is computationally impractical. Variance reduction is therefore necessary.

## 5.4 Variance reduction via geometric control variate

The geometric Asian option uses a geometric rather than an arithmetic mean of the monitoring prices. While not actively traded in markets, it has one property of substantial practical value: the geometric average of correlated lognormals is itself lognormal. Consequently the geometric Asian admits a closed-form pricing formula, derived in Kemna and Vorst (1990).

On each simulated price path the engine computes both payoffs in parallel: the arithmetic payoff  $Y_i$  (the quantity of interest) and the geometric payoff  $X_i$  (the auxiliary quantity with known mean). Because both payoffs are functions of the same simulated path, they are strongly correlated. Under the running configuration this correlation is approximately 0.99.

This high correlation underpins the control variate estimator:

$$\hat{V}_{cv} = \frac{1}{N} \sum_{i=1}^N [Y_i - \beta(X_i - \mu_X)]$$

Where:

- $Y_i$  is the discounted arithmetic Asian payoff on path  $i$
- $X_i$  is the discounted geometric Asian payoff on the same path
- $\mu_X$  is the true price of the geometric Asian, computed from the Kemna-Vorst formula
- $\beta$  is a small constant chosen to make the error as tight as possible

The construction works as follows. On any individual path, if the arithmetic estimate  $Y_i$  is high by random chance, the geometric estimate  $X_i$  on the same path is also high. The term  $(X_i - \mu_X)$  therefore captures the path-specific random error, and subtracting it from  $Y_i$  removes most of the sampling noise. The estimator remains unbiased because  $\mu_X$  is known exactly, so the subtraction does not shift the sample mean.

Under the running configuration, the standard error is reduced from 0.036 (plain Monte Carlo) to 0.00106 (with control variate), corresponding to a 34-fold reduction in standard error and a 1,138-fold reduction in variance. Achieving comparable accuracy with plain Monte Carlo would require approximately 57 million paths.

## 6. Results for the running example

### 6.1 Price summary

Table 1 reports the price of the option together with four supporting statistics evaluated under the running configuration.

| Quantity                | Value            | Interpretation                                                                     |
|-------------------------|------------------|------------------------------------------------------------------------------------|
| Option Price            | 5.7828           | Fair value of the call option today, in units of the underlying.                   |
| 95% Confidence Interval | [5.7807, 5.7848] | Statistical interval containing the true price with 95% probability.               |
| Standard Error          | 0.00106          | Sampling error of the Monte Carlo estimate; smaller values imply higher precision. |
| Runtime                 | $\approx 200$ ms | Wall-clock time to simulate 50,000 paths and apply the control variate.            |
| Variance Reduction      | 1,138×           | Ratio of plain Monte Carlo variance to control variate variance.                   |

**Table 1:** Headline pricing results under the running configuration.

Under this configuration the option is at the money, so its intrinsic value is zero and the entire premium of 5.78 corresponds to time value.

### 6.2 Risk sensitivities (Greeks)

The five standard Greeks are the first-order sensitivities of the option price to each input. They quantify the local response of the price to a small change in a single parameter and form the basis of any hedging strategy. Their values under the running configuration are summarised in Table 2.

| Greek          | Value   | Interpretation                                                                   |
|----------------|---------|----------------------------------------------------------------------------------|
| $\Delta$ Delta | +0.5889 | A unit increase in spot price raises the option price by approximately 0.59.     |
| $\Gamma$ Gamma | +0.0325 | A unit increase in spot raises Delta by 0.032; measures price curvature in spot. |
| $\nu$ Vega     | +0.2179 | A 1 percentage-point rise in volatility raises the option price by 0.22.         |
| $\Theta$ Theta | -0.0093 | The option loses approximately 0.0093 units of value per calendar day.           |
| $\rho$ Rho     | +0.2459 | A 1 percentage-point rise in the risk-free rate raises the price by 0.25.        |

**Table 2:** The five standard Greeks under the running configuration.

The signs are all consistent with theoretical predictions:

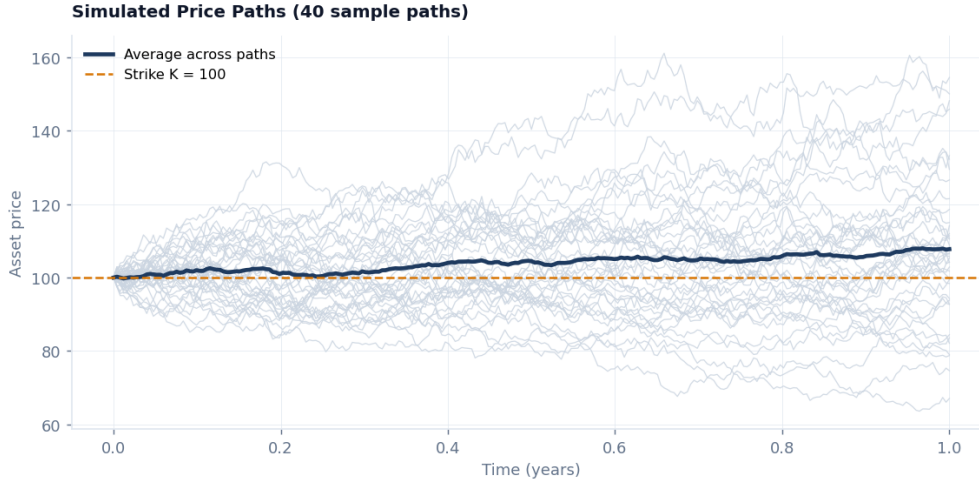
- Call Delta lies in  $[0, 1]$ .
- Gamma is positive: the option price is convex in spot.
- Vega is positive: greater volatility raises optionality value.
- Theta is negative: the option decays in value as time passes.

- Rho is positive for a call option.

As an internal consistency check, Delta is computed by two independent methods (finite differences with common random numbers, and the pathwise estimator of Broadie and Glasserman, 1996), and the two estimates agree within their joint confidence interval.

### 6.3 Visualisation of simulated paths

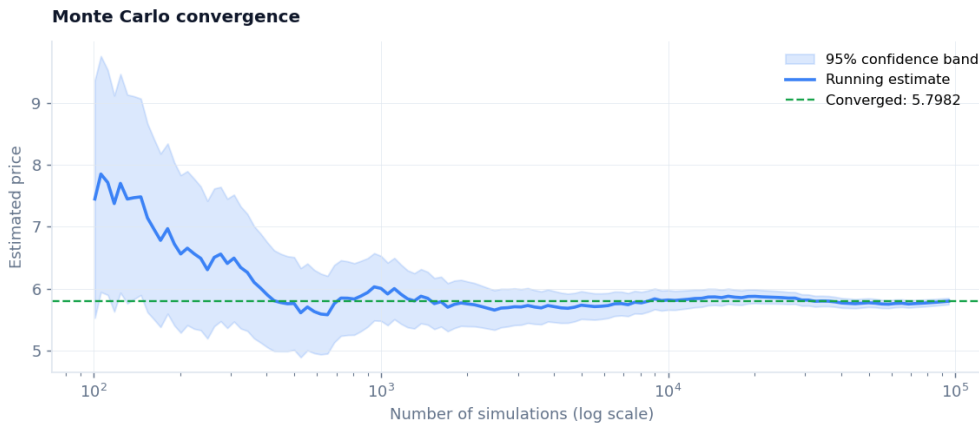
Figure 1 displays 40 simulated paths of the underlying asset price. The grey lines correspond to the individual paths and the navy line to their average across all 40. The amber dashed line marks the strike level  $K = 100$ .



**Figure 1:** Forty simulated price paths under the GBM model. Two paths can end at the same final price but have very different averages, which is why the Asian payoff depends on the entire shape of the journey.

### 6.4 Convergence of the Monte Carlo estimator

Figure 2 reports the behaviour of the price estimate as the number of simulated paths increases. The horizontal axis is the path count on a logarithmic scale, ranging from 100 to 100,000. The blue line traces the running estimate, and the shaded band shows the corresponding 95% confidence interval. For small  $N$  (below approximately 1,000) the estimate exhibits substantial fluctuation. As  $N$  approaches and exceeds 50,000, the running estimate stabilises in a narrow neighbourhood of the converged value (5.80, marked by the green dashed line). This provides empirical confirmation that 50,000 paths is sufficient for the chosen target accuracy.



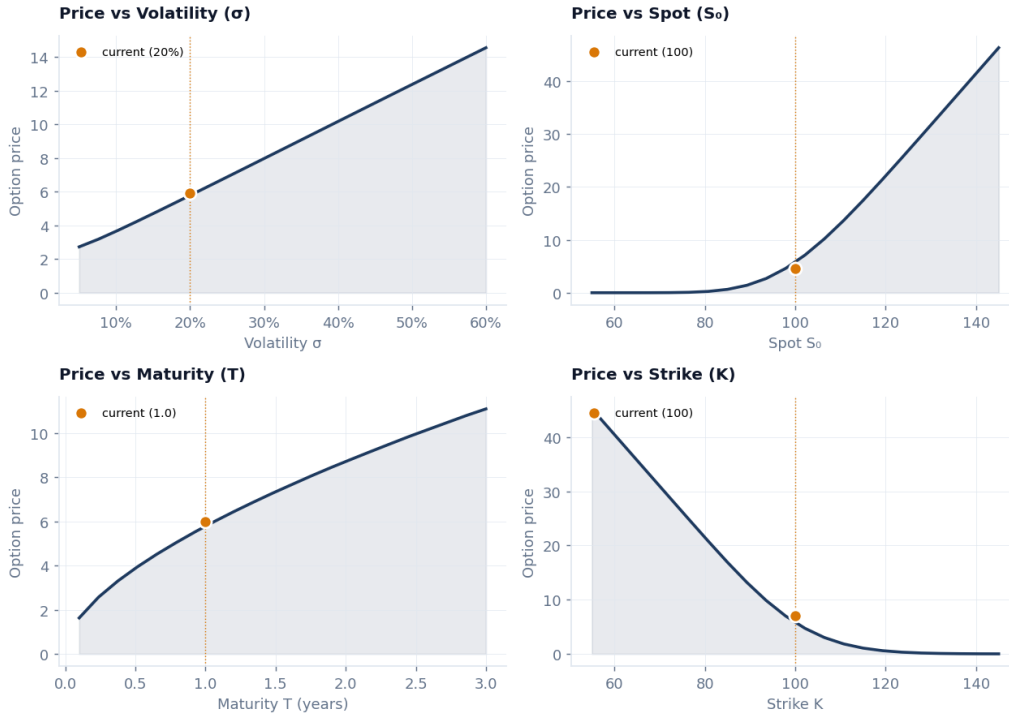
**Figure 2:** The running estimate (blue line) settles down toward the true price as more paths are simulated. The confidence band shrinks at the rate  $1/\sqrt{N}$ .

## 7. Sensitivity analysis

The sensitivity analysis quantifies the response of the option price to changes in each input parameter, addressing the requirement in the project brief that the report describe how the price varies when parameter values are perturbed.

### 7.1 Univariate sensitivities

Figure 3 reports the option price as a function of each of the four primary input parameters (volatility, spot, maturity, and strike). In each panel the parameter on the horizontal axis is varied over a realistic range while the remaining three are held at their default values. The marker indicates the position of the running example.



**Figure 3:** Price as a function of each input, holding the other three at the defaults. The amber dot marks the running example.

**Volatility ( $\sigma$ ).** The price is monotonically increasing in volatility. Higher  $\sigma$  widens the distribution of possible averages, which raises the expected value of the right-skewed payoff  $\max(\bar{A} - K, 0)$ . Within the range tested, the relationship is approximately linear.

**Spot ( $S_0$ ).** The price exhibits the classical S-shape with respect to spot. For values well below the strike, the option is essentially worthless. For values well above the strike, the price approaches  $S_0 - Ke^{-rT}$ , the intrinsic value of an in-the-money position. At the money, where the running example is located, the curve has its steepest slope.

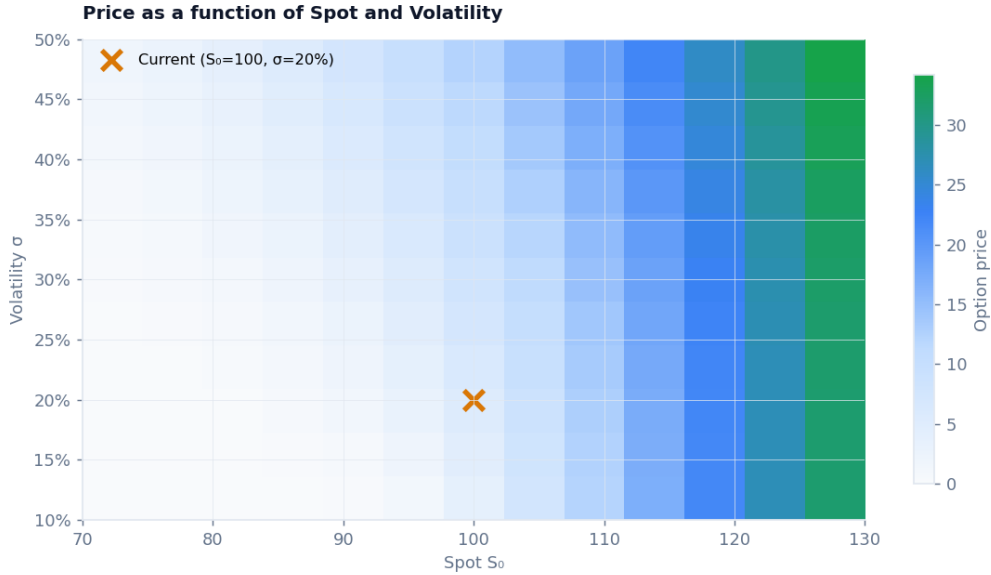
**Maturity ( $T$ ).** The price is increasing and concave in maturity, since the standard deviation of the underlying scales with  $\sqrt{T}$  rather than  $T$ . The concavity becomes visible at maturities beyond one year.

**Strike ( $K$ ).** A higher strike reduces the call payoff and hence the price. The relationship is convex, reflecting the bounded downside of the payoff.

### 7.2 Joint sensitivity: Spot–Volatility surface

The univariate panels above each hold three inputs at their default values. Figure 4 examines the joint effect of varying spot and volatility simultaneously. The price at each (spot, volatility) pair is encoded as a colour intensity.

The maximum price is attained in the upper-right corner, corresponding to a combination of deep

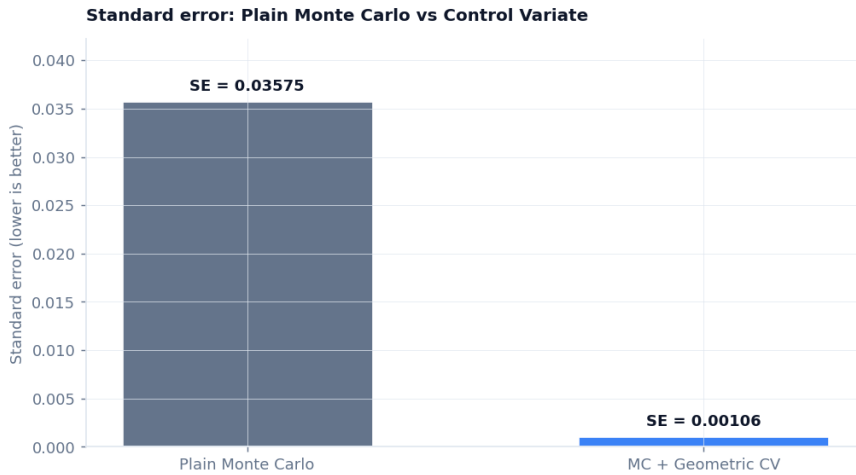


**Figure 4:** Joint sensitivity to spot and volatility. The amber cross marks the running example.

in-the-money spot and high volatility. The amber cross marks the running example, which lies in a near-neutral region of the surface. A simultaneous upward shock in spot and volatility (the conditions characteristic of a bullish news event in an already volatile market) would shift the operating point along the steepest gradient toward the upper-right corner.

## 8. Variance reduction result

The principal numerical contribution of the project is quantified in this section. Figure 5 compares the standard error of the plain Monte Carlo estimator against that of the control variate estimator, holding the number of simulated paths fixed at  $N = 50,000$ .



**Figure 5:** Standard error of the price estimate, plain Monte Carlo versus the control variate version. Both use 50,000 paths.

| Method                         | Price         | Standard Error | Variance Reduction |
|--------------------------------|---------------|----------------|--------------------|
| Plain Monte Carlo              | 5.7740        | 0.03575        | 1.0×               |
| MC + Geometric Control Variate | <b>5.7828</b> | <b>0.00106</b> | <b>1,138×</b>      |

Two observations follow:

- Both methods produce the same price estimate, as expected. The control variate construction is unbiased and therefore cannot shift the central estimate.
- The standard error of the control variate estimator is approximately 34 times smaller than that of

the plain estimator. Since variance scales with the square of the standard error, this corresponds to the reported 1,138-fold variance reduction. Matching this precision with plain Monte Carlo would require on the order of 57 million paths.

## 9. Conclusion

The main findings of the project are summarised as follows.

- The arithmetic Asian option, an exotic contract with no closed-form solution, was priced using Monte Carlo simulation under risk-neutral Geometric Brownian Motion dynamics.
- Variance was reduced by approximately a factor of 1,138 through the geometric-Asian control variate based on the Kemna and Vorst (1990) closed-form formula.
- For the running example (at-the-money,  $\sigma = 20\%$ ,  $T = 1$  year), the engine returned a call price of 5.7828 with a 95% confidence interval of [5.7807, 5.7848] and a runtime of approximately 200 milliseconds.
- All five Greeks (Delta, Gamma, Vega, Theta, Rho) exhibited signs and magnitudes consistent with theoretical predictions and were cross-validated against an independent pathwise estimator for Delta.
- The methodology assumes constant volatility, no jumps, a flat risk-free rate, and no transaction costs; relaxing these assumptions (volatility smile, jump-diffusion, yield curve) constitutes the natural next stage of work.

## References

1. Kemna, A. G. Z. and Vorst, A. C. F. (1990). A pricing method for options based on average asset values. *Journal of Banking and Finance*, 14, 113–129.
2. Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer.
3. Hull, J. C. (2018). *Options, Futures, and Other Derivatives*, 10th edition. Pearson.
4. Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
5. Broadie, M. and Glasserman, P. (1996). Estimating security price derivatives using simulation. *Management Science*, 42(2), 269–285.